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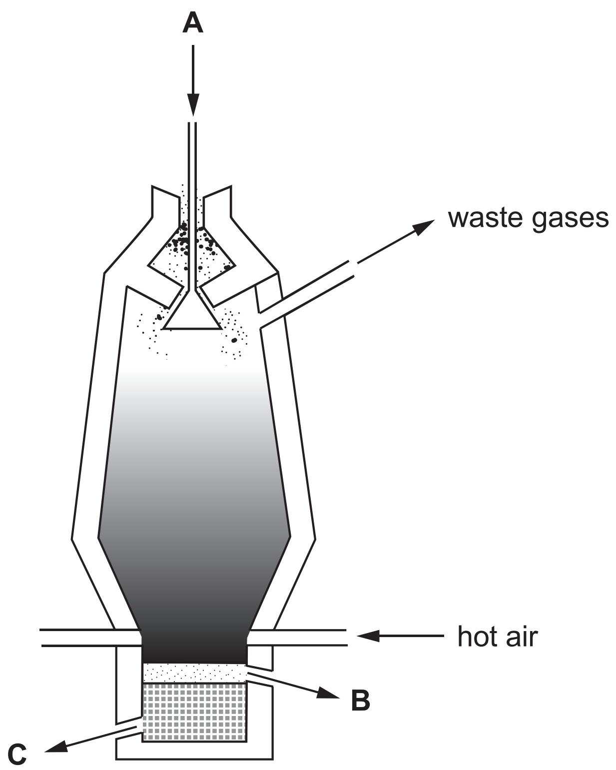
Answer **all** *questions*.

1. *(a)* The diagram shows where substances enter and leave the blast furnace in the extraction of iron from iron ore.



coke	**slag**	**iron**	**limestone**	**iron ore**

(i) Use the substances in the box to complete the following sentences. [3]

The **three** raw materials which enter the furnace at **A** are .... ,

.... and ....

Product **B** is .... and product **C** is ....

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(ii) One reaction that takes place in the furnace is

iron(III) oxide + carbon monoxide $\longrightarrow$ iron + carbon dioxide

Underline the element which is removed from the iron(III) oxide during the reaction. [1]

iron **oxygen** **carbon**

(b) Statements **D**, **E** and **F** show the three steps needed to prepare a sample of copper(II) chloride in the laboratory. The steps are **not** in the correct order.

D filter to remove excess copper(II) oxide

E leave the copper(II) chloride solution to evaporate at room temperature

F add excess copper(II) oxide to dilute hydrochloric acid

Complete the flow chart by putting the **letters** in the correct order. [2]

☐ $\longrightarrow$ ☐ $\longrightarrow$ ☐

first step last step

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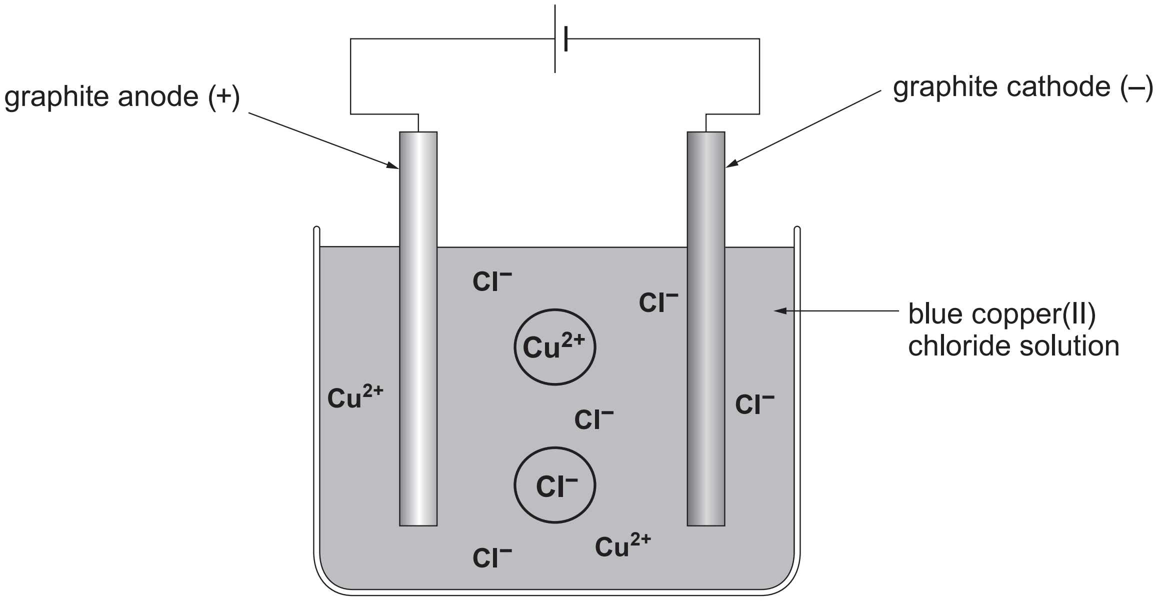
(c) The diagram shows the apparatus used to obtain copper from copper(II) chloride solution.



electrolyte	**electrodes**	**electricity**	**electrolysis**

(i) Choose words from the box to complete the following sentences. [2]

The breaking down of copper(II) chloride solution using an electric current is called

....

Electric current enters and leaves the copper(II) chloride solution by the

....

(ii) **By drawing arrows on the diagram**, show the direction in which the **circled** ions move during the process. [1]

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(iii) The reaction occurring at the cathode is:

$$\underset{\text{copper(II) ion}}{Cu^{2+}} + 2e^{-} \longrightarrow \underset{\text{copper atom}}{Cu}$$

I. Put a tick (✓) in the box next to what e^{-} represents in the equation. [1]

electricity ☐

electron ☐

ion ☐

atom ☐

II. Copper(II) ions, Cu^{2+}, are removed from the solution during the process.

Put a tick (✓) next to the statement which describes what, if anything, happens to the colour of the copper(II) chloride solution. [1]

the blue solution turns darker ☐

the blue solution turns paler ☐

the blue solution turns yellow ☐

nothing happens to the colour of the blue solution ☐

(iv) Put a tick (✓) next to the gas which is formed at the anode. [1]

oxygen ☐

hydrogen ☐

chlorine ☐

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